

EPC2361 Half-Bridge Analytical Loss Modeling

Updated operating point, loss equations and waveforms

19 September 2026 — Implementation: *epc2361_loss.py* and *waveform_plotter.py*

1 Operating point and scope

The model uses one EPC2361 in each of the two switch positions. The defaults are a 75 V DC bus, 40 A AC RMS current, 100 kHz switching frequency, sinusoidal PWM, **modulation index** $m = 1$, and unity power factor. The waveform factor $k_{\text{wf}} = 1$ scales the sinusoidal current amplitude; it does not model arbitrary harmonics. The assumed half bridge has one hard turn-on/off pair per PWM period, with natural commutation of the opposite device. This is a first-order estimate, not a DC-load model.

The retained circuit assumptions are $1\ \Omega$ external gate resistance in each direction, $0.5\ \Omega$ source/sink driver resistance, 2 A driver current limits, 50 ns effective dead time, and 2 nH loop inductance. The default case temperature is 50 °C and ambient temperature is 25 °C.

$$I_{\text{pk}} = \sqrt{2} I_{\text{rms}} k_{\text{wf}} = 56.57\ \text{A}, \quad I_{\text{avg,abs}} = \frac{2I_{\text{pk}}}{\pi} = 36.01\ \text{A}.$$

RMS current determines conduction loss; mean absolute current determines average switching and dead-time loss. The plotted event defaults to peak current.

2 Conduction loss and junction temperature

The default `rds_temperature_mode=iterative` couples conduction loss to junction temperature. The maximum 25 °C resistance, 1.0 m Ω , is selected by default. At each iteration the normalized datasheet curve gives

$$\begin{aligned} R_{\text{DS,on}}(T_J) &= R_{\text{DS,on},25} k_T(T_J), \\ d_{\text{dead}} &= 2t_{\text{dead,eff}} f_{\text{sw}} = 0.01, \\ P_{\text{cond}}(T_J) &= I_{\text{rms}}^2 k_{\text{wf}}^2 R_{\text{DS,on}}(T_J) (1 - d_{\text{dead}}), \\ T_J^{(n+1)} &= T_{\text{ref}} + \frac{R_{\theta}}{2} [P_{\text{cond}}(T_J^{(n)}) + P_{\text{overlap}} + P_{\text{COSS}} + P_{\text{dead}}]. \end{aligned}$$

The code starts at the selected case or ambient temperature and repeats until the thermal residual is below 10^{-6} °C, with a maximum of 200 iterations. This numerical tolerance does not imply equivalent physical accuracy.

Piecewise-linear interpolation uses approximate readings of datasheet Figure 9 [1]. At 0, 25, 50, 75, 100, 125 and 150 °C, the respective normalized factors are 0.84, 1.00, 1.16, 1.33, 1.49, 1.65 and 1.80. The typical normalized curve scales the selected typical or maximum 25 °C resistance; it is not a guaranteed hot-resistance limit. No extrapolation is made outside 0–150 °C. Leaving that range or failing to converge produces an explicit error instead of a reported steady-state solution.

The converged defaults are $T_J = 50.4922\ \text{°C}$, $R_{\text{DS,on}} = 1.16335\ \text{m}\Omega$, and $P_{\text{cond}} = 1.84274\ \text{W}$ for both FETs combined. The small dead-time correction avoids double counting normal conduction during reverse-conduction intervals. Optional external loop loss is $P_{\text{loop}} = I_{\text{rms}}^2 k_{\text{wf}}^2 R_{\text{loop}}$.

Manual mode remains available and uses the entered multiplier (default 1.7) without feedback. Iterative mode ignores that manual multiplier. The result reports updated resistance, junction temperature, iteration count and residual. It estimates steady-state temperature under the specified cooling conditions; it does not measure actual junction temperature.

3 Device data and transition timing

The supplied EPC2361 datasheet [1] gives typical/maximum on-resistance of 0.75/1.0 m Ω , typical threshold voltage 1.1 V, internal gate resistance 0.4 Ω , and zero reverse-recovery charge. At 50 V and 50 A, the gate-charge values are $Q_G = 28$ nC, $Q_{GS} = 8.5$ nC, $Q_{G(th)} = 6$ nC and $Q_{GD} = 3.8$ nC. Following AN030 [2],

$$Q_{GS2} = Q_{GS} - Q_{G(th)} = 2.5 \text{ nC}.$$

The code uses an approximate plateau voltage $V_{PL} = 2.1$ V from Figure 7. At 75 V, integrating the small additional C_{RSS} between 50 and 75 V gives approximately $Q_{GD} = 4.0$ nC and $Q_G = 28.2$ nC. These are nominal curve-based estimates. Plateau and current-transition charge remain fixed near the 50 A characterization condition; exact current and temperature dependence is not solved.

With 5 V turn-on and 0 V turn-off drive, define

$$\begin{aligned} R_{on} &= R_{G,on} + R_{\text{driver,source}} + R_{G,int} = 1.9 \text{ } \Omega, \\ R_{off} &= R_{G,off} + R_{\text{driver,sink}} + R_{G,int} = 1.9 \text{ } \Omega, \\ \bar{V}_G &= (V_{th} + V_{PL})/2. \end{aligned}$$

The gate-current estimates are separately limited by driver capability:

$$\begin{aligned} I_{G,CR} &= \min\left(\frac{V_{DR} - \bar{V}_G}{R_{on}}, I_{\text{source}}\right), & I_{G,VF} &= \min\left(\frac{V_{DR} - V_{PL}}{R_{on}}, I_{\text{source}}\right), \\ I_{G,VR} &= \min\left(\frac{V_{PL}}{R_{off}}, I_{\text{sink}}\right), & I_{G,CF} &= \min\left(\frac{\bar{V}_G}{R_{off}}, I_{\text{sink}}\right). \end{aligned}$$

The resulting current-rise, voltage-fall, voltage-rise and current-fall intervals are

$$\begin{aligned} t_{CR} &= Q_{GS2}/I_{G,CR} = 1.40 \text{ ns}, & t_{VF} &= Q_{GD}/I_{G,VF} = 2.62 \text{ ns}, \\ t_{VR} &= Q_{GD}/I_{G,VR} = 3.62 \text{ ns}, & t_{CF} &= Q_{GS2}/I_{G,CF} = 2.97 \text{ ns}. \end{aligned}$$

Threshold and post-plateau segments use segment-average gate voltage and current limits to complete the illustrative gate waveform. They are not exact nonlinear RC transients, and the gate-charge segments must not exceed total Q_G .

4 Switching overlap loss

For an event at current I_{event} , the triangular overlap approximation [2] is

$$\begin{aligned} E_{on} &= \frac{1}{2} V_{\text{bus}} I_{\text{event}} (t_{CR} + t_{VF}), \\ E_{off} &= \frac{1}{2} V_{\text{bus}} I_{\text{event}} (t_{VR} + t_{CF}). \end{aligned}$$

At 56.57 A these are 8.52 and 13.97 μJ . The line-cycle average instead uses $I_{\text{avg,abs}}$:

$$P_{\text{overlap}} = \frac{1}{2} V_{\text{bus}} I_{\text{avg,abs}} (t_{CR} + t_{VF} + t_{VR} + t_{CF}) f_{\text{sw}} = 1.4323 \text{ W}.$$

Peak-current event energy is not applied to every PWM cycle.

5 Capacitor, gate-drive and dead-time losses

For a symmetric half bridge, AN030 p6, Eqs17–18 [2], gives

$$P_{\text{COSS}} = V_{\text{bus}} Q_{\text{OSS}} (V_{\text{bus}}) f_{\text{sw}} = 0.8550 \text{ W}.$$

Datasheet Figure 6 gives approximately $Q_{\text{OSS}} = 114 \text{ nC}$ and $E_{\text{OSS}} = 3.2 \text{ } \mu\text{J}$ at 75 V. The energy is reported for reference, not added again. Neither small-signal C_{OSS} nor Q_{OSS}/V is an energy-equivalent capacitance for a nonlinear device. When bus voltage changes, update the charge and energy inputs to that voltage; mismatched charge-reference voltage is rejected.

The total gate-supply loss for the two FETs is

$$P_{\text{gate}} = 2Q_G V_{\text{DR}} f_{\text{sw}} = 0.0282 \text{ W}.$$

This energy is dissipated across the driver and gate resistances. Driver quiescent, bootstrap and isolated-supply overhead are not included.

Using an approximate reverse drop $V_{\text{SD}} = 2.2 \text{ V}$ from datasheet Figure 8,

$$P_{\text{dead}} = 2V_{\text{SD}} I_{\text{avg,abs}} t_{\text{dead,eff}} f_{\text{sw}} = 0.7923 \text{ W}.$$

The 1.6 V datasheet value applies at only 0.5 A. The selected 2.2 V is a representative estimate at the modeled currents, not a fixed guaranteed drop. Effective dead time may differ from commanded dead time in either direction [3]. Treating the entire selected interval as reverse conduction is conservative: capacitor self-commutation and duty-dependent distortion are not resolved. No reverse-recovery term is added.

6 Total loss, efficiency and thermal result

$$\begin{aligned} P_{\text{loss}} &= P_{\text{cond}} + P_{\text{overlap}} + P_{\text{COSS}} + P_{\text{gate}} + P_{\text{dead}} + P_{\text{loop}} \\ &= 4.9505 \text{ W}, \\ V_{\text{out,rms}} &= \frac{mV_{\text{bus}}}{2\sqrt{2}} = 26.5165 \text{ V}, \\ P_{\text{out}} &= V_{\text{out,rms}} I_{\text{rms}} \text{PF} = 1060.66 \text{ W}, \\ \eta &= 100 \frac{P_{\text{out}}}{P_{\text{out}} + P_{\text{loss}}} = 99.5354\%. \end{aligned}$$

These efficiency figures include only the modeled losses, not all converter/system losses. Output voltage uses the retained sinusoidal half-bridge convention.

The thermal calculation excludes external loop and gate-supply losses:

$$\begin{aligned} P_{\text{semi}} &= P_{\text{cond}} + P_{\text{overlap}} + P_{\text{COSS}} + P_{\text{dead}}, \\ P_{\text{device}} &= P_{\text{semi}}/2 = 2.46114 \text{ W}, \\ T_J &= T_{\text{case}} + P_{\text{device}} R_{\theta JC} = 50.4922 \text{ } ^\circ\text{C}. \end{aligned}$$

Equal sharing applies to symmetric sinusoidal averaging. The default $R_{\theta JC} = 0.2 \text{ } ^\circ\text{C/W}$ assumes the case is maintained at 50 °C. Ambient mode instead uses $T_J = T_{\text{amb}} + P_{\text{device}} R_{\theta JA}$ with 44 °C/W for the JEDEC board (25 °C/W for the datasheet EVB). The small internal-gate-resistance share is omitted from junction heating. Conduction loss is recomputed with the converged resistance. The other loss terms remain fixed during iteration; temperature dependence of gate charge, reverse drop and switching energy is not modeled.

7 Waveforms, slew rates and voltage stress

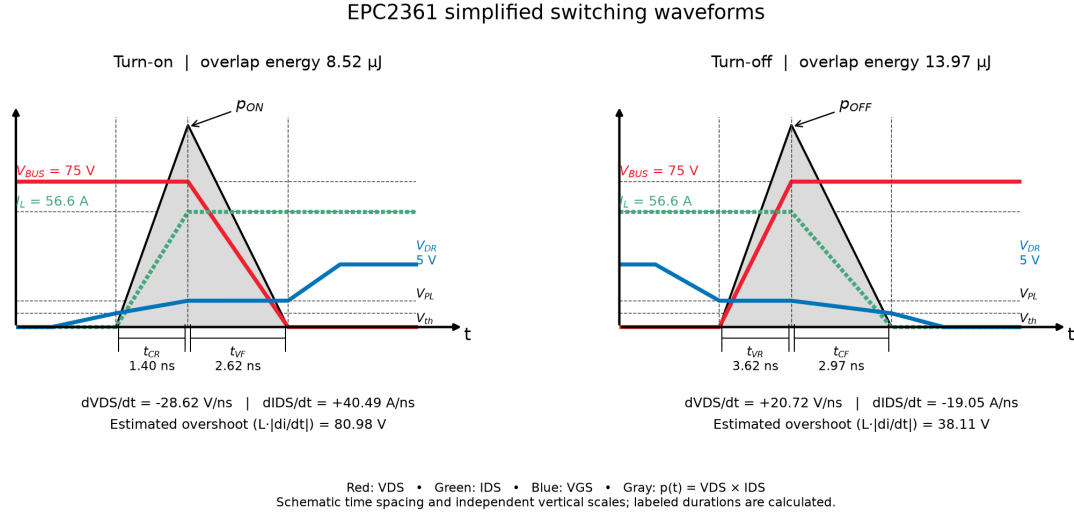


Figure 1. Simplified turn-on and turn-off waveforms following AN030 Figures 2–3. Time spacing and vertical scales are schematic; the labeled intervals and slew rates are calculated. Gray shading represents channel overlap power. The physical-time waveform arrays integrate to the reported event energies.

The signed slopes refer to V_{DS} and I_{DS} in their respective transition intervals. The overshoot indicators use their current-slope magnitudes:

$$\Delta V_{on} = L_{loop} \frac{I_{event}}{t_{CR}} = 80.98\text{ V},$$

$$\Delta V_{off} = L_{loop} \frac{I_{event}}{t_{CF}} = 38.11\text{ V},$$

$$V_{stress} = V_{bus} + \max(\Delta V_{on}, \Delta V_{off}) = 155.98\text{ V}.$$

This crude estimate exceeds the device’s 100 V continuous rating. The retained 2 nH input is exposed rather than adjusted to suppress the result. The calculation does not solve nonlinear commutation, common-source inductance, damping or gate feedback, so it is not a prediction of a measured voltage peak. Validate timing and stress with a parasitic-aware model and double-pulse measurements [4].

Disabling hard switching suppresses both overlap and capacitor dissipation as an ideal fully soft-switched limit; it does not establish ZVS. Optional LC ringing is illustrative and does not change the analytical loss accounting. Thermal transients, PCB/heatsink spreading, dynamic on-resistance and driver overhead remain outside the model. Ten numerical checks cover loss accounting, waveform integration, input handling, thermal balance, an independent closed-form solution and failure conditions.

References

- [1] EPC, *EPC2361 Datasheet*, revised 20 July 2026: pp1–2 for ratings and charge data; p3, Figures5–9 for capacitance, charge, reverse conduction and temperature.
- [2] EPC, AN030, *Hard Switching Losses Calculation*: pp2–5 for gate-charge transitions and overlap; p6 for capacitor and gate-drive loss.
- [3] EPC, *Dead-Time Optimization for Maximum Efficiency*, p1.
- [4] EPC, *eGaN FET Drivers and Layout Considerations; Impact of Parasitics on Performance*; AN023, *Accurately Measuring High Speed GaN Transistors*; *Thermal Performance of eGaN FETs*.